# Behaviour Transitions and Metabolic Change in a 3-Year Health Check-Up Cohort

Re-analysis of „Effect of a 2-hour interval [...]" (10.1136/bmjnph-2018-000011 )

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*This re-analysis does not represent the perspective of initial or current authors, and is intended only as exemplary tutorial material.*

EASD European Association for the Study of Diabetes

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# On the Maw et al. 2019 study

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- Population: Japanese adults aged 40–74 years from City A, Okayama Prefecture, who underwent specific health examinations.
- Exposure: Sleeping within 2 hours after dinner at least 3 times per week.
- Comparison: More than 2 hours between dinner and bedtime.
- Outcome: Change in glycated haemoglobin (HbA1c) over time.

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Brief methods: Analysis of repeated annual health check-up data from 2012, 2013, and 2014 with **22-item self-report lifestyle questionnaire**, anthropometric and laboratory data.

Stats:

1) **Univariate analyses**,
2) **GLMMs**: time of measurement as level 1 and individual-level predictors as level 2.
   Covariates: **BMI, triglycerides, blood pressure, smoking, alcohol use, and regular exercise**.

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Brief results:

1) Shorter dinner-to-bedtime interval did not significantly affect HbA1c trajectory over time.
2) HbA1c levels in general were rising.
3) Other factors (BMI, smoking, alcohol consumption) were better predictors of HbA1c change over time.

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https://www.bbc.co.uk/programmes/b03z9jr0

# Health-Promoting Behaviors

**Modifiable factors**

## Dietary Habits

Meal timing, breakfast consumption, and eating speed

## Physical Activity and Sleep

Regular physical activity and sleep length and regularity

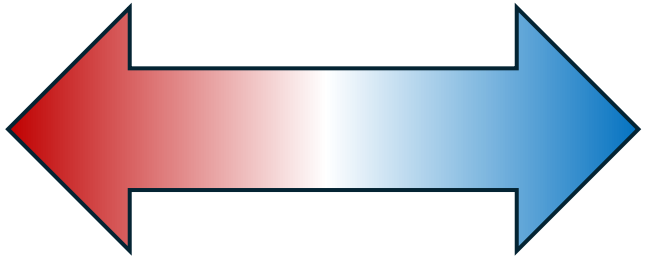
## Metabolic Effects

Adiposity, glycemic control, and cardiovascular risk.

# Behavioral change and glycemic progression

## Prediabetes risk

Poor dietary habits (late meals, excessive alcohol) might increase the risk of prediabetes.



## Return to normoglycemia

Beneficial lifestyle changes increase the chance of returning to normal glycemia.

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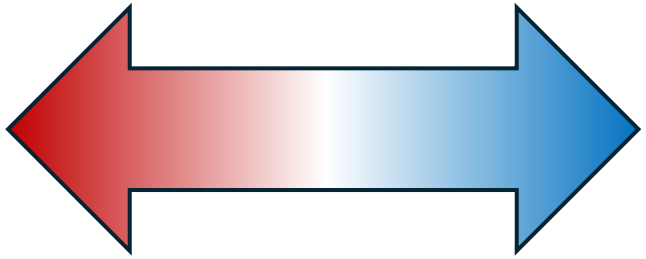

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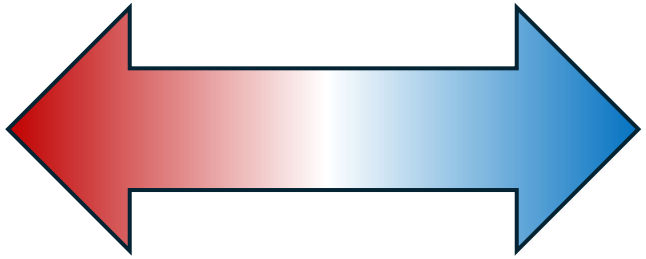

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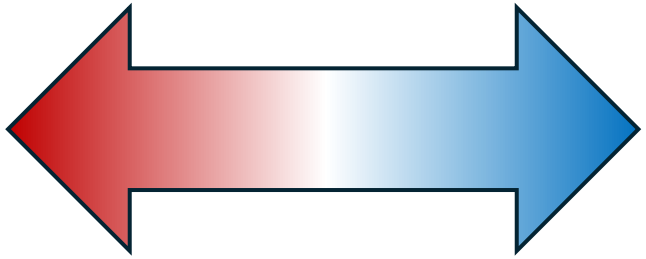

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*Are trends observed at populational level still true at the intra-individual level?*

*We could run an RCT and spend millions. Issue: adherence, self-selection*

*Alternatively, we can search for signals in public data and spend days. Issue: unreliable reporting, recall bias.*

The NEW ENGLAND JOURNAL of MEDICINE



EDITORIAL

# Data Sharing

**Authors**: Dan L. Longo, M.D., and Jeffrey M. Drazen, M.D. Author Info & Affiliations

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The aerial view of the concept of data sharing is beautiful. What could be better than having high-quality information carefully reexamined for the possibility that new nuggets of useful data are lying there, previously unseen? The potential for leveraging existing results for even more benefit pays appropriate increased tribute to the patients who put themselves at risk to generate the data. The moral imperative to honor their collective sacrifice is the trump card that takes this trick.

However, many of us who have actually conducted clinical research, managed clinical studies and data collection and analysis, and curated data sets have concerns about the details. The first concern is that someone not involved in the generation and collection of the data may not understand the choices made in defining the parameters. Special problems arise if data are to be combined from independent studies and considered comparable. How heterogeneous were the study populations? Were the eligibility criteria the same? Can it be assumed that the differences in study populations, data collection and analysis, and treatments, both protocol-specified and unspecified, can be ignored?

A second concern held by some is that a new class of research person will emerge — people who had nothing to do with the design and execution of the study but use another group's data for their own ends, possibly stealing from the research productivity planned by the data gatherers, or even use the data to try to disprove what the original investigators had posited. There is concern among some front-line researchers that the system will be taken over by what some researchers have characterized as "research parasites."

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https://researchparasite.com/

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# On our team's 2026 re-analysis

- Population: Japanese adults aged 40–74 years from City A, Okayama Prefecture, excluding HbA1c>6.5%, with <2 timepoints.
- Exposure: Behaviour change for quick eating, eating within 2hr before bed, midnight snack, skipping breakfast.
- Comparison: No behaviour change.
- Outcome: Change in waist circumference, lipids, HbA1c, BMI (after adjusting for baseline outcome level, age, sex, smoking, alcohol drinking, exercise, medication use, and baseline BMI and/or HbA1c where appropriate).

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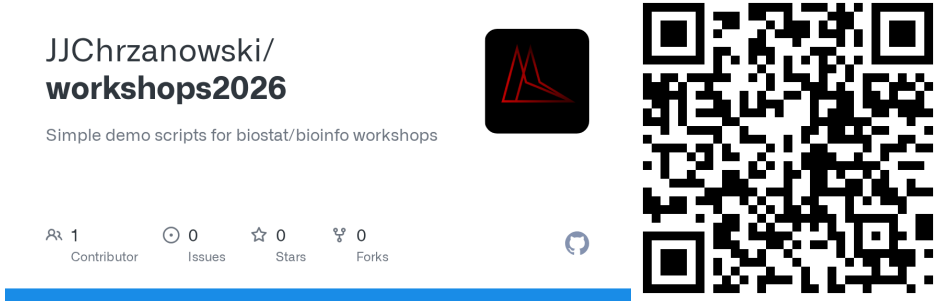
Stats:

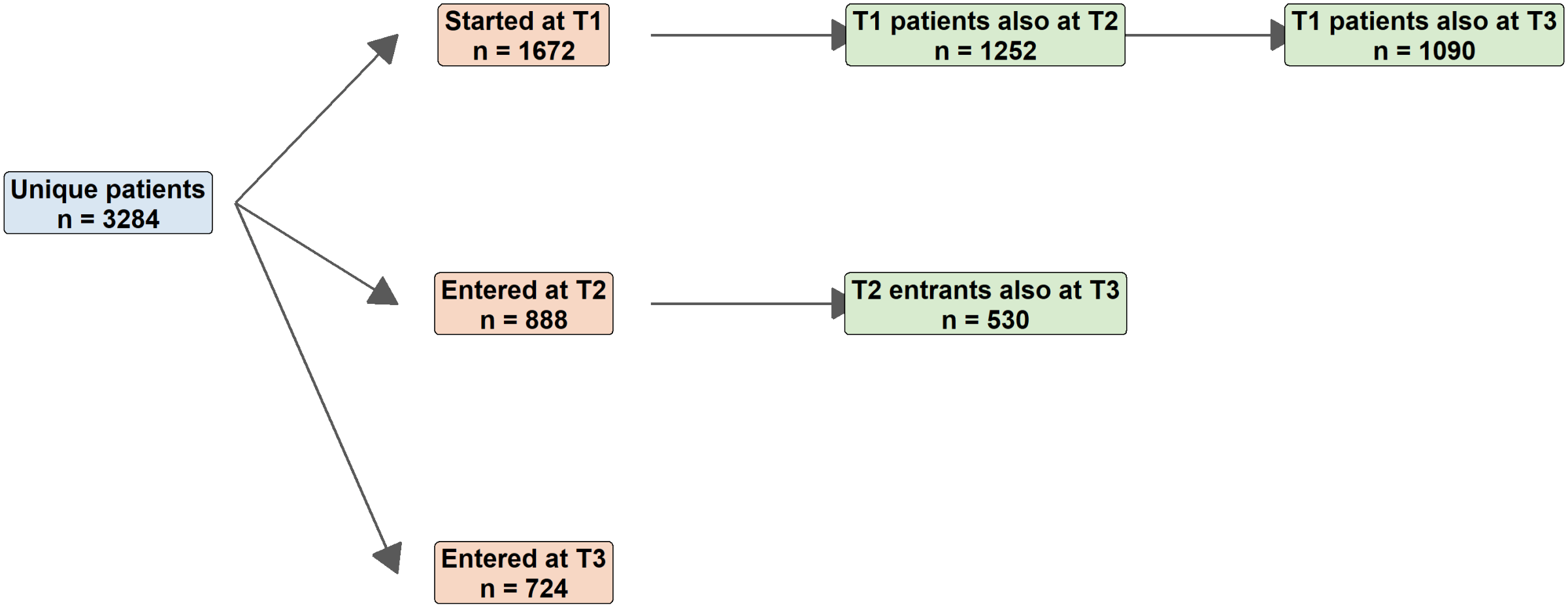
1) **Alluvian plot visualization for behaviour change**,
2) **Linear regression model for changes** in:
    1) waist circumference,
    2) Lipids (TG, LDL, HDL),
    3) HbA1c,
    4) BMI.

    Covariates: **baseline outcome level, age, sex, smoking, alcohol drinking, exercise, medication use, and cohort entry timepoint**.

To ensure reproducibility, the code for this reanalysis is available on GitHub.



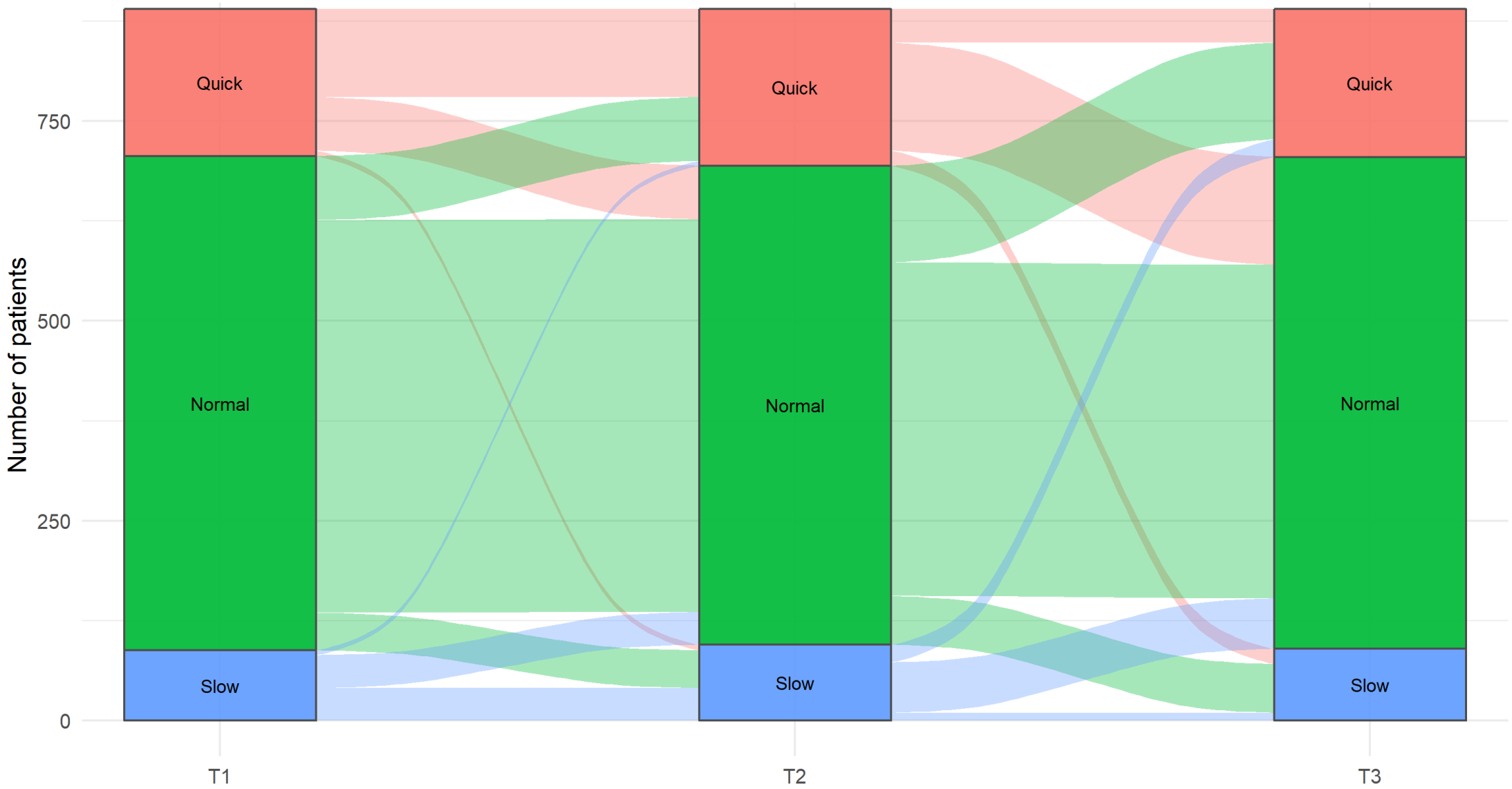
# Patient flow across timepoints

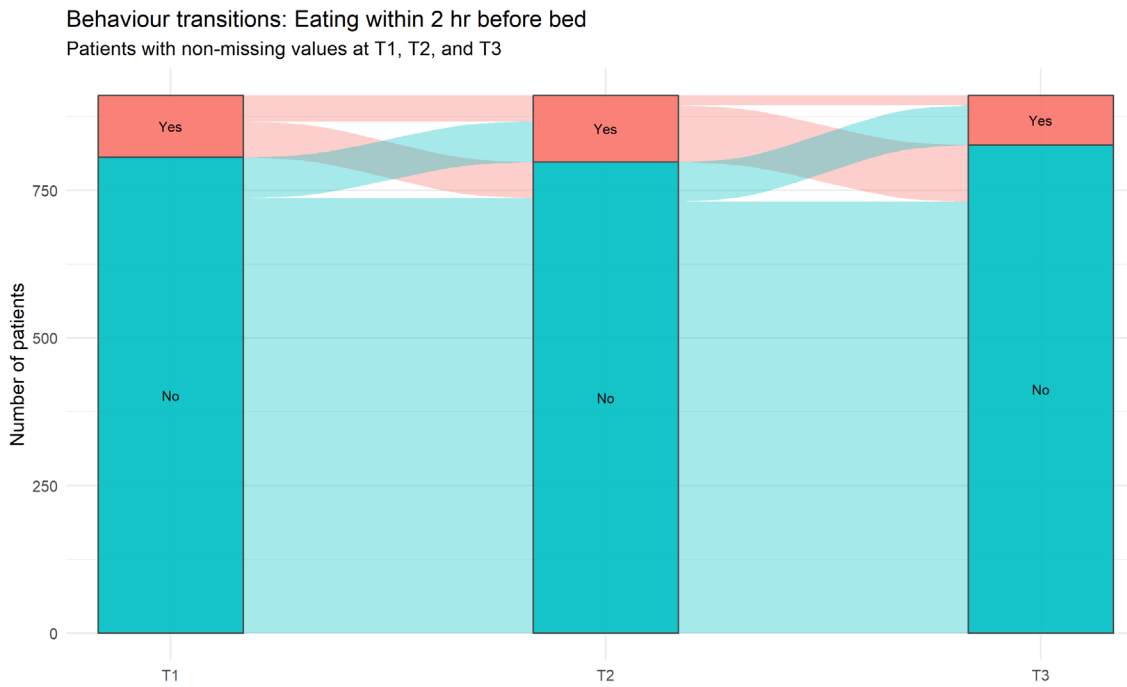


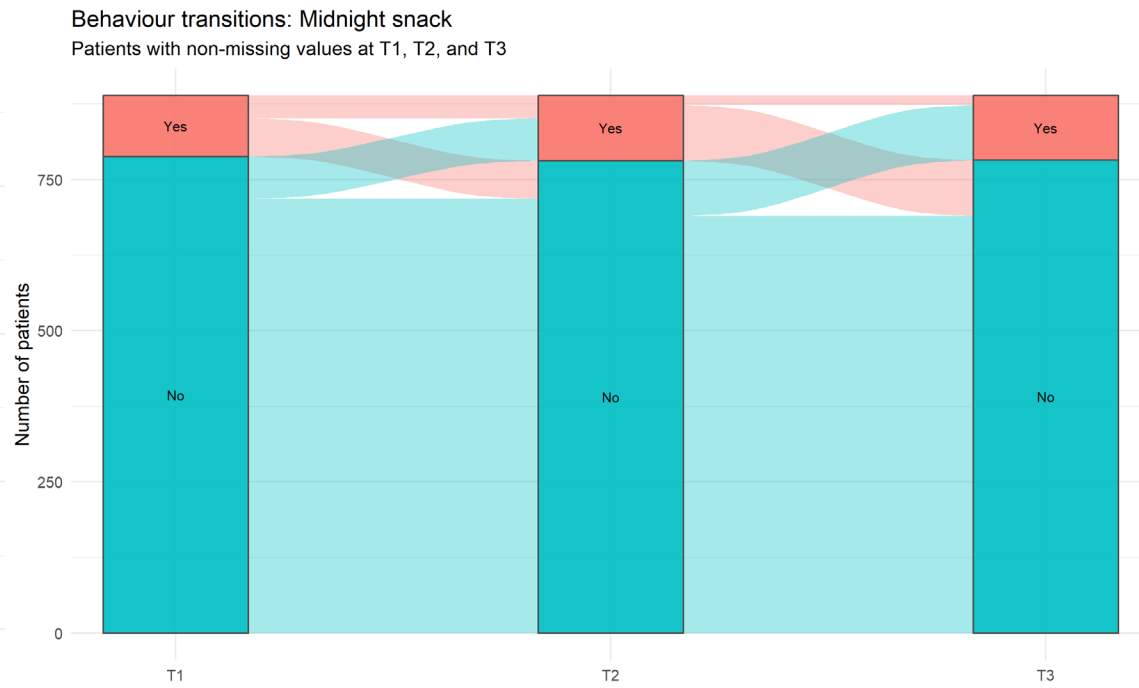
# General characteristics

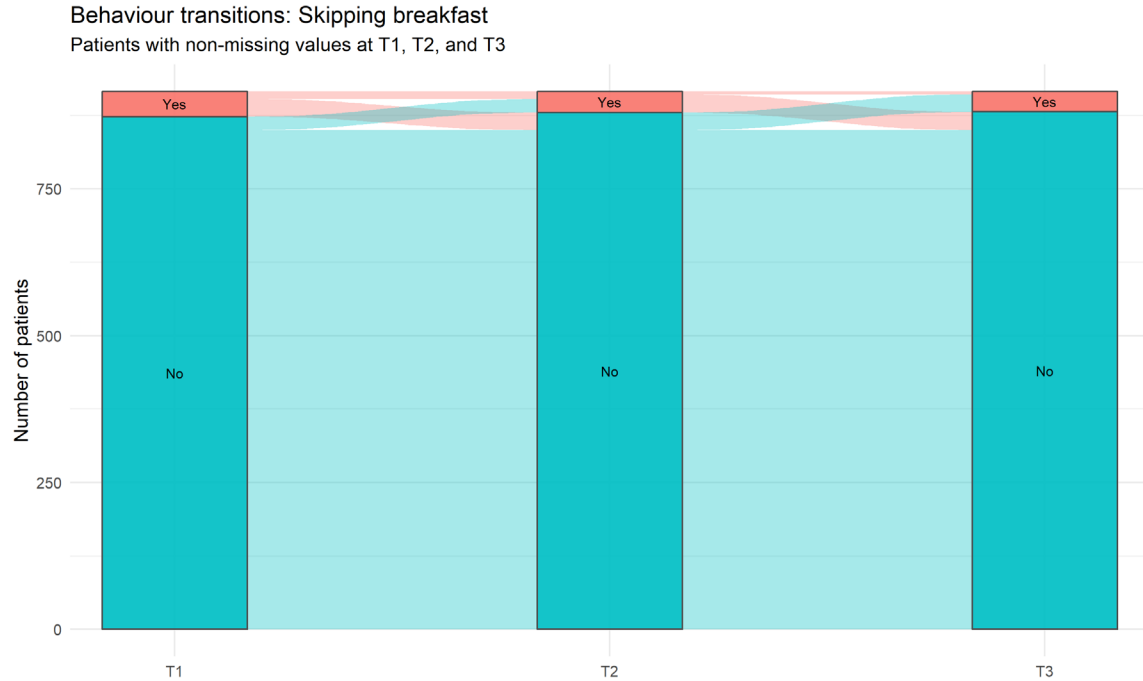
| | First timepoint |
|---|---|
| **Patients** | 1895 |
| **Enrolled at first study year** | 1365 (72%) |
| **Age, years** | 64.9±7.4 |
| **Male** | 712 (37.6%) |
| **HbA1c, %** | 5.41±0.57 |
| **BMI, kg/m^2** | 23.0±3.4 |
| **Waist circumference, cm** | 82.9±9.5 |
| **LDL, mg/dL** | 127.4±31.5 |
| **Triglyceride, mg/dL** | 102 [75, 140] |
| **Current smoking** | 191 (10.1%) |
| **Hypertension medication** | 623 (32.9%) |
| **Hyperlipidemia medication** | 473 (2.5%) |
| **Diabetes medication** | 187 (9.9%) |

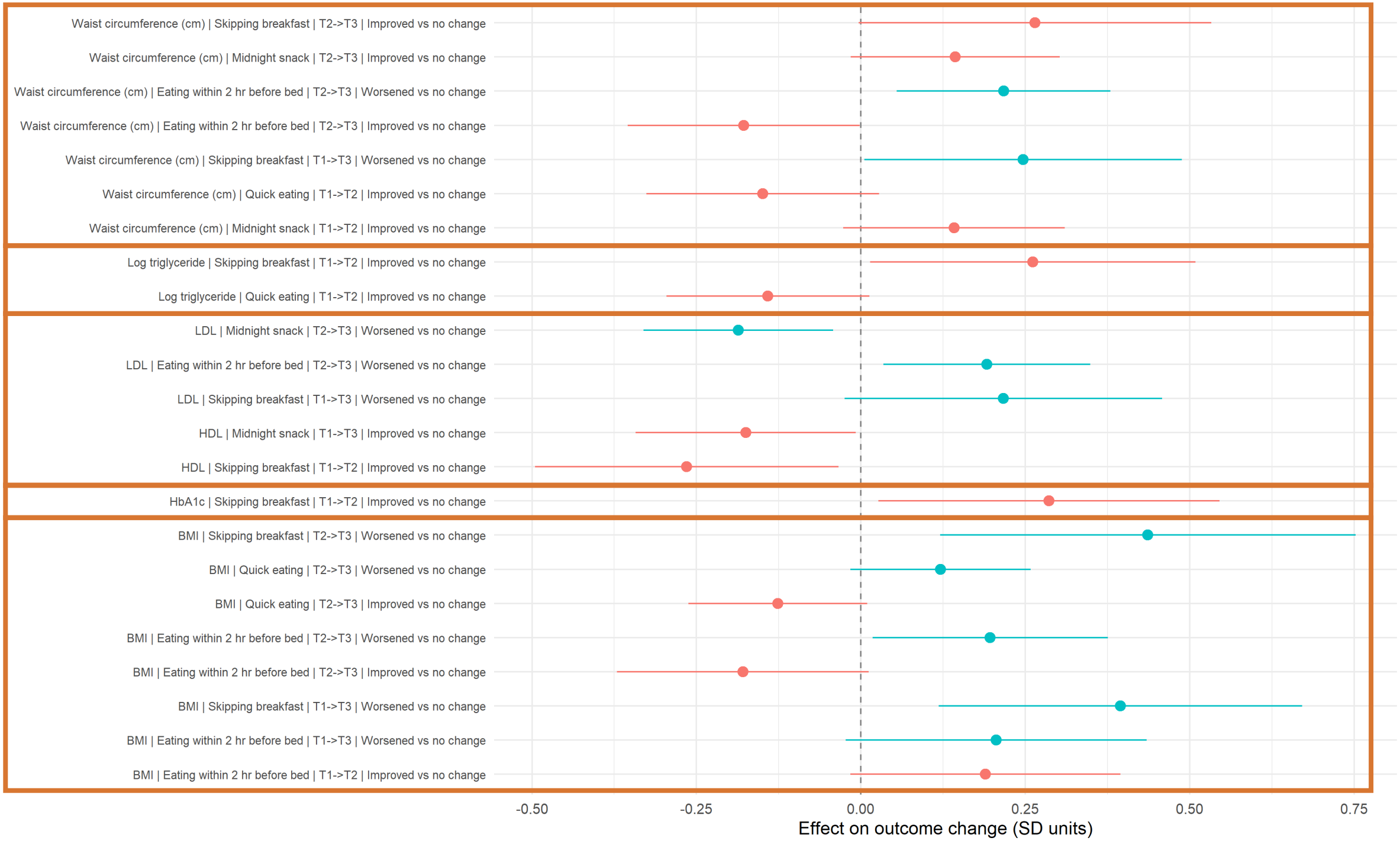
# Behaviour transitions – Quick eating




Behaviour transitions: Eating within 2 hr before bed
Patients with non-missing values at T1, T2, and T3
Number of patients
0
250
500
750
Yes
No
T1
T2
T3


Behaviour transitions: Midnight snack
Patients with non-missing values at T1, T2, and T3
Number of patients
0
250
500
750
Yes
No
T1
T2
T3


Behaviour transitions: Skipping breakfast
Patients with non-missing values at T1, T2, and T3
Number of patients
0
250
500
750
Yes
No
T1
T2
T3


Waist circumference (cm) | Skipping breakfast | T2->T3 | Improved vs no change
Waist circumference (cm) | Midnight snack | T2->T3 | Improved vs no change
Waist circumference (cm) | Eating within 2 hr before bed | T2->T3 | Worsened vs no change
Waist circumference (cm) | Eating within 2 hr before bed | T2->T3 | Improved vs no change
Waist circumference (cm) | Skipping breakfast | T1->T3 | Worsened vs no change
Waist circumference (cm) | Quick eating | T1->T2 | Improved vs no change
Waist circumference (cm) | Midnight snack | T1->T2 | Improved vs no change
Log triglyceride | Skipping breakfast | T1->T2 | Improved vs no change
Log triglyceride | Quick eating | T1->T2 | Improved vs no change
LDL | Midnight snack | T2->T3 | Worsened vs no change
LDL | Eating within 2 hr before bed | T2->T3 | Worsened vs no change
LDL | Skipping breakfast | T1->T3 | Worsened vs no change
HDL | Midnight snack | T1->T3 | Improved vs no change
HDL | Skipping breakfast | T1->T2 | Improved vs no change
HbA1c | Skipping breakfast | T1->T2 | Improved vs no change
BMI | Skipping breakfast | T2->T3 | Worsened vs no change
BMI | Quick eating | T2->T3 | Worsened vs no change
BMI | Quick eating | T2->T3 | Improved vs no change
BMI | Eating within 2 hr before bed | T2->T3 | Worsened vs no change
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BMI | Skipping breakfast | T1->T3 | Worsened vs no change
BMI | Eating within 2 hr before bed | T1->T3 | Worsened vs no change
BMI | Eating within 2 hr before bed | T1->T2 | Improved vs no change
-0.50
-0.25
0.00
0.25
0.50
0.75
Effect on outcome change (SD units)

# Study limitations

- Observational design **does not establish causality**.
- Self-reported behaviors may be subject to recall bias.
- Loss to follow-up due to diabetes development?

# Conclusions

- Behaviour changes are mostly reflected in BMI and waist trajectories.
- Associations with HbA1c and lipids are weaker and inconsistent.
- Transition-based analysis reveals patterns otherwise missed by baseline-only approaches.

Take-home messages:

- Behaviour change matters!
- There is a lot of data out there – do not be afraid to explore it.
- Statistics and cats are fun.